

TSMC

Margin test passed; On to 2026 with strong AI and N2 growth, with upside to GMs from pricing

The strongest message from TSMC's 2Q25 earnings conference was the resilience of GMs, despite unprecedentedly steep TWD appreciation and increased overseas Fab dilution. We expect 4Q25 to be the GM trough (due to UTR draw-down), with GMs likely to rebound in 2026 with price hikes, better N3 yield and accelerated productivity and cost-down measures. Demand trends for AI are strong with broader demand pickup (CSPs, sovereign AI and China demand) and CoWoS capacity likely to remain tight into 2026. While TSMC is cautious on 4Q25 (we now model 6% qoq decline in revenues), chances are that demand could shape up better if tariffs turn out to be less intense than feared. As such, we believe TSMC could eventually finish the year with low-mid 30% yoy growth in USD terms (JPMe 32%). We continue to see healthy growth in 2026, with a faster N2 ramp, price hikes from 1Q26 and continued strength in N3 due to strong AI accelerator demand. We raise our FY25/ FY26 EPS estimates by 5/4% to reflect stronger growth prospects in AI and stay OW with a Dec-25 PT of NT\$1,275. In the near term, we believe the stock may stay rangebound as the market is likely to focus on Sec 232 and specific country tariff outcomes. Once some clarity emerges on this front, we believe the stock should start pricing in strong growth and GM rebound in 2026, with some prospect of meaningful N2 outsourcing from Intel.

- Passing the GM test; 2026 should look better:** TSMC's 3Q Gross Margin guidance (55.5-57.5%, JPMe 56.7%) is quite impressive, considering a 600 bps drag from TWD appreciation and 100-200bps from overseas expansion since 1Q25. Increased utilization has helped, but strong cost-cutting measures have also contributed to keeping costs under check. But for the steep TWD appreciation, we believe TSMC could have easily reached 60+% GM levels by now, demonstrating underlying pricing power and operational excellence. We believe that 4Q25 GMs are likely to also be down qoq, dragged down by lower utilization (due to the qoq decline in revenues). But 4Q25 should be the near-term bottom for GMs, with price hikes and a better N3 mix kicking in from early 2026. We expect GMs to remain higher than 2H25 levels through 2026, despite some dilution in late 2026 as N2 starts ramping up.
- Datacenter AI growth looks stronger; continued growth in 2026 with N3 migration kicking in:** TSMC indicated that Datacenter AI demand growth appears stronger with strong token growth at hyperscalers, upside from sovereign AI demand and likely upside materializing in a few months from China, after H20 resumption, in our view. We expect Datacenter AI growth expectations to be revised up again (from current mid 40% CAGR into 2029). Looking into 2026, we expect 50-60% growth in Datacenter AI, helped by strong growth in CoWoS capacity for AI accelerators, better pricing due to migration of most accelerators from N4 to N3 and incremental contribution from HBM base-dies. We believe that CoWoS demand expectations are likely to be raised again (after a couple of rounds of cuts in 1H25), especially if NVDA starts orders H20 and B20 related products for the China market.
- Pricing uptick to come through: Double-digit blended ASP increases are**

Overweight

2330.TW, 2330 TT
Price (17 Jul 25):NT\$1130.0
Price Target (Dec-25):NT\$1275.0

Technology and Telecoms

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Key Changes (FYE Dec)

	Prev	Cur
Adj. EPS - 25E (NT\$)	55.47	58.44
Adj. EPS - 26E (NT\$)	62.68	65.11

Quarterly Forecasts (FYE Dec)

Adj. EPS (NT\$)	2024A	2025E	2026E
Q1	8.70	13.94A	13.88
Q2	9.56	15.36A	15.77
Q3	12.54	15.39	17.23
Q4	14.45	13.75	18.24
FY	45.24	58.44	65.11

Style Exposure

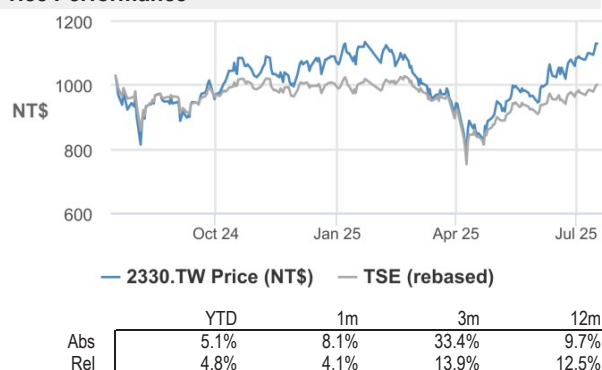
Quant Factors	Current %Rank	Hist %Rank (1=Top)			
		6M	1Y	3Y	5Y
Value	40	45	54	65	54
Growth	20	15	19	16	35
Momentum	36	11	15	38	42
Quality	13	15	10	18	20
Low Vol	39	43	31	27	29
ESGQ	24	92	4	11	1

Sources for: Style Exposure – J.P. Morgan Quantitative and Derivatives Strategy; all other tables are company data and J.P. Morgan estimates.

See page 21 for analyst certification and important disclosures, including non-US analyst disclosures.

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Price Performance



Company Data

Shares O/S (mn)	25,930
52-week range (NT\$)	1160.0-780.0
Market cap (\$ mn)	996,034
Exchange rate	29.42
Free float (%)	71.5%
3M ADV (mn)	34.62
3M ADV (\$ mn)	1,169.1
Volatility (90 Day)	41
Index	TAIEX
BBG ANR (Buy Hold Sell)	40 2 0

Key Metrics (FYE Dec)

NT\$ in millions	FY24A	FY25E	FY26E	FY27E
Financial Estimates				
Revenue	2,894,308	3,618,960	4,118,088	4,634,792
Adj. EBIT	1,322,053	1,710,940	1,942,311	2,175,018
Adj. EBITDA	1,984,850	2,451,520	2,857,399	3,201,001
Adj. net income	1,173,268	1,515,408	1,688,455	1,902,176
Adj. EPS	45.24	58.44	65.11	73.35
BBG EPS	44.92	58.34	67.09	78.45
Cashflow from operations	1,975,661	1,785,853	2,620,930	2,928,958
FCFF	1,142,360	561,038	1,315,930	1,754,458
Margins and Growth				
Revenue Growth Y/Y (%)	33.9%	25.0%	13.8%	12.5%
EBIT margin	45.7%	47.3%	47.2%	46.9%
EBIT Growth Y/Y (%)	43.5%	29.4%	13.5%	12.0%
EBITDA margin	68.6%	67.7%	69.4%	69.1%
EBITDA Growth Y/Y (%)	36.5%	23.5%	16.6%	12.0%
Net margin	40.5%	41.9%	41.0%	41.0%
Adj. EPS growth	39.9%	29.2%	11.4%	12.7%
Ratios				
Adj. tax rate	16.6%	16.4%	16.2%	16.3%
Interest cover	NM	NM	NM	NM
Net debt/Equity	NM	NM	NM	NM
Net debt/EBITDA	NM	NM	NM	NM
ROE	30.3%	31.7%	29.1%	27.8%
Valuation				
FCFF yield	3.9%	1.9%	4.5%	6.0%
Dividend yield	1.5%	1.8%	2.2%	2.9%
EV/Revenue	9.7	7.7	6.6	5.7
EV/EBITDA	14.1	11.4	9.5	8.2
Adj. P/E	25.0	19.3	17.4	15.4

Summary Investment Thesis and Valuation

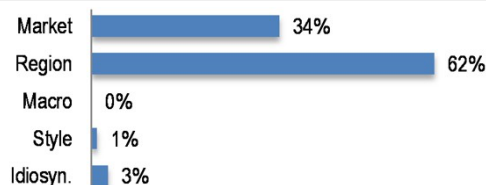
Investment Thesis

We expect TSMC's structural growth drivers to remain strong due to its near-monopoly position in AI accelerators and edge AI, supported by its robust process roadmap (N3 and N2) and industry-leading packaging technology. We believe TSMC could grow low-mid 30% in USD terms in 2025 and healthy growth in 2026, driven by broader AI demand pick-up (e.g. CSPs, sovereigns and China demand) with CoWoS capacity remaining tight into 2026. Additionally, we anticipate the GM to rebound in 2026 with price hikes, better N3 yield, and accelerated productivity and cost-down measures.

Valuation

Our Dec-25 PT of NT\$1,275 is based on ~20x 12-month forward P/E and reflects the positive fundamental demand driver, continued growth in N3 and stronger N2 ramp in 2026. Our target multiple is higher than TSMC's five-year average historical multiple.

Performance Drivers



Factors	6M Corr	1Y Corr
Market: MSCI Asia Pac ex JP	0.59	0.58
Region: Taiwan	0.97	0.97
Macro:		
Emerging Central Bank Rate	-0.01	-0.15
JPM Global Equity Sentiment	0.23	0.14
HSI Volatility Index	0.17	-0.12
Quant Styles:		
Growth	0.34	0.54
DivYld	-0.15	-0.34
Value	-0.16	-0.34

now the norm: As highlighted before, we believe that TSMC is now in a new period of growth, where double-digit blended ASP growth is the norm. Hence, 5-6% wafer shipment growth should help drive ~20% growth in revenues, which should also drive capital intensity lower and help drive strong 30+% ROEs over the long-term. For 2026, we continue to expect leading edge wafer prices to move up again as TSMC reflects increased costs, TWD appreciation, increased overseas mix, etc into the wafer prices. We believe N3 is likely to see the most upside, followed by N2, N4 and certain forms of CoWoS. Overall, we see 6-10% price increases across the leading edge process nodes.

- **N2 ramp appears faster and smoother; Broad smartphone and HPC adoption, with upside risk from Intel:** We expect N2 to ramp to be 60-65k wfpw of output by end 2026, a faster ramp than N3 and should contribute \$6.8bn of revenues in 2026 (~2x of first year revenues for N3). Yield ramp for N2 also appears quite smooth with initial chip probing yield already reaching 70% for early products, based on our research. Apple appears increasingly likely to use N2 for all four new iPhones in 2H26, while Android flagship SoCs from QCOM and MediaTek are also likely to adopt N2. We also see small wafer demand for AMD and Intel for their N2 products. Looking into 2027, we believe all leading edge customers except NVDA are likely to be shipping N2 based products - a much faster adoption curve compared to N3 or N5. Intel remains the primary upside risk - we believe that Intel is yet to sign a capacity agreement with TSMC for N2, but this looks increasingly likely, given the scale down of Intel 18A (likely only 1 phase of 25-30k wfpw, well below the normal peak for a leading edge node). We anticipate that Intel's client PC products are likely to be outsourced to TSMC N2 in 2027-28 timeframe.
- **Near-term eyes largely on tariffs and the degree of weakness in 4Q:** Investor attention is likely to be largely focused on the implied qoq weakness in 4Q. We believe that TSMC is taking a conservative view on tariffs (Sec 232 and country-specific tariffs) and is not factoring in any exemptions for TSMC or Taiwan at this point. TSMC also appears to be anticipating a marked slowdown in consumer demand as a result (including iPhone, in our view). We are still modeling a 6% qoq decline in 4Q25, but feel that such a sharp slowdown in demand may not materialize, given that tariff outcomes so far have been better than feared. Consumer demand (esp for smartphones and PCs) has also been less price sensitive in recent years, given that these markets have already been ex-growth for many years. We do believe that there has been some pull-in demand in consumer, PC etc. in 1H25, but the extent is likely to be small given the ongoing tightness in N5 and N3 capacity since 2H24. Overall, we believe that TSMC's 4Q25 revenues could come in better than guidance, with FY25 ending up with low 30% USD growth.
- **Capex outlook mixed, but we expect upside in 2026; Strong, multiyear dividend growth ahead:** TSMC's commentary on capex was more mixed, and did not indicate continued growth in capex into 2026, despite very strong N2 demand. We expect 2026 capex to grow to \$45bn, driven by strong N2 demand, continued strength in N3 and continued accelerated buildout of capacity in the US (tools move in for the 2nd AZ Fab and construction of 3rd and 4th AZ Fab in 2026). We also expect N2 demand to be upsized in case Intel signs a capacity agreement with TSMC for large-scale N2 outsourcing. We believe that TSMC's cash dividends should grow steadily due to strong FCF generation, and expect a 2023-27 DPS CAGR of 26%, with 2027E dividends reaching NT\$33 levels.

Passing the GM test; 2026 should look better

TSMC reported a stronger-than-expected 2Q GM of 58.6%, beating both JPMe and Consensus by 0.7-1ppt and reaching the high-end of its guidance (57-59%) despite unfavorable FX (around 125bps impact) thanks to higher-than-expected UTR and cost improvement efforts.

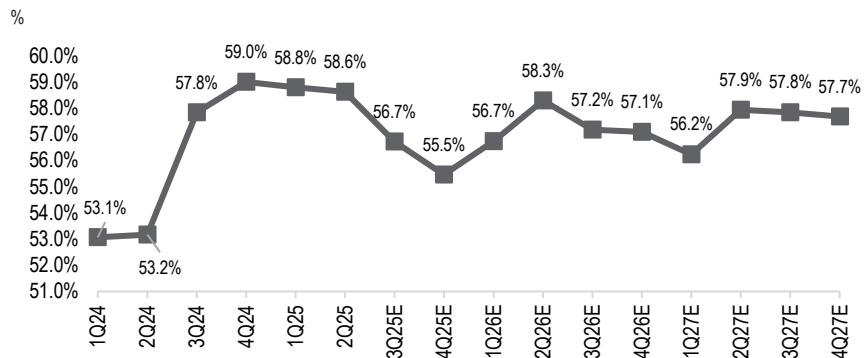
Company guided 3Q rev to grow 8% qoq at mid point, with GM likely at 55.5-57.5% (down 2.1ppts qoq at mid point). The margin guidance is quite impressive in our view, considering a 600 bps drag from TWD and 100-200bps from overseas expansion since 1Q25.

While 4Q GM would likely further decline with the UTR draw down, we expect 4Q25 to be the near-term bottom for GMs.

2H25 should be the period with maximum stress on GMs (fastest period of TWD appreciation so far in 3Q25, overseas cost dilution, some negate demand impact dragging utilization lower in 4Q25 with price hikes not kicking in yet). Even in this period, TSMC appears to be able to hold GMs well above 53%, demonstrating its underlying pricing power and operational efficiency. Without the recent TWD appreciation, we believe TSMC would have been able to achieve 60+% GMs at this point.

With price hikes and a better N3 mix kicking in from early 2026, we expect GMs to remain higher than 2H25 levels through 2026, despite some dilution in late 2026 as N2 starts ramping up.

Figure 1: TSMC Quarterly GM



Source: Company data, J.P. Morgan estimates.

Datacenter AI growth looks stronger; continued growth in 2026 with N3 migration kicking in

TSMC expressed confidence in the growth of datacenter AI, driven by strong token growth from hyperscalers (the top 4 US CPS capex to grow 38%/20% YoY in 2025/2026), robust sovereign AI demand and increased demand from China following the resumption of H20 (projected at 1 million units per quarter initially, see our note, [link](#)).

We anticipate that AI growth expectations are likely to be revised up in the following quarters due to sustained AI demand and potential upside from China (driven by lifting of H20 restrictions). In addition, management believes that the supply-demand gap for CoWoS will persist into 2026, exceeding investors' expectations of a balanced CoWoS demand by that year.

As Apple and other flagship smartphone manufacturers transition from N3 to N2 in 2026, we expect many AI accelerators to shift to the N3P and N3X processes starting in late 2025, with additional contributions from HBM base-die revenues once HBM4 adoption increases in 2H26. We project N3 utilization to remain above 100% in 2026, with ASP for N3 increasing by 19% YoY (price hike and better mix due to higher HPC share), contributing to a 35% YoY revenue growth for N3 in 2026. Overall, we anticipate 50-60% growth in datacenter AI revenue in 2026.

Figure 2: Timeline for Datacenter AI accelerators migration to 3 nm

Company	Product / Project	2024				2025				2026				2027			
		1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q
Nvidia	B300A																
	Rubin																
AMD	MI325																
	MI350																
Google	TPU v6e																
	TPU v6p																
Amazon	Trainium 2.5 (Cayman)																
	Trainium 3 (Mariana)																
Microsoft	MAIA 1																
	MAIA 2-2																
META	MTIA 2																
	MTIA 3 (Iris)																

Source: J.P. Morgan estimates, Company data.

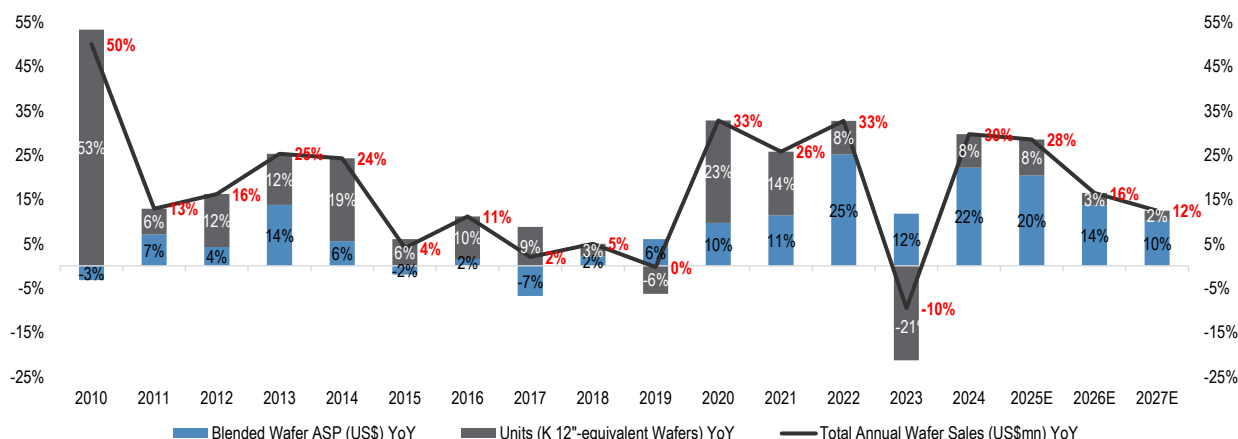
Pricing uptick coming through; Double-digit blended ASP increases now the norm

TSMC has now seen more than 10 quarters of double-digit YoY ASP increases, including a 16% YoY rise in 2Q25. Given strong pricing power and intention to raise pricing to reflect increasing costs and technology leadership, we believe that low-mid teens blended ASP growth will become the norm for TSMC moving forward. As a result, a 5-6% increase in wafer shipments should contribute to ~20% revenue growth, which in turn should likely reduce capital intensity and support strong 30+% ROE over the long term.

With TSMC indicating that capacity remains tight, particularly for N5 and N3, and a faster ramp-up of N2, we anticipate pricing to rise in 2026 (6-10% increase in leading-edge wafer prices, along with higher pricing for overseas fabs) to offset cost increases, some impact of TWD appreciation and overseas expansion.

Overall, for 2026, we believe that N3 will see the strongest price increases (due to strong AI accelerator demand and tight supply), followed by N2, N4 and CoWoS.

Figure 3: Breakdown of TSMC wafer sales by blended ASP and wafer shipments



Source: J.P. Morgan estimates, Company data.

N2 ramp appears faster and smoother; Broad smartphone and HPC adoption, with upside risk from Intel

We've mentioned in our previous notes ([link](#), [link](#)) that N2 progress appears to be quite strong. In the first 15 months of N2 ramp, we see AMD (ramping in late 2025), Apple, Mediatek, Qualcomm and Intel (likely ramping in 2026) all starting their N2 wafers at TSMC.

Apple appears increasingly likely to use N2 for all four new iPhones in 2H26, while Android flagship SoCs from QCOM and Mediatek should also adopt N2. We also see small wafer demand for AMD and Intel for their N2 products. Looking into 2027, we believe all leading edge customers except NVDA are likely to be shipping N2 based products - a much faster adoption curve compared to N3 or N5.

Intel remains as an upside risk. While the capacity agreement with TSMC for N2 has not yet signed, this looks increasingly likely given the scale down of Intel 18A (likely only 1 phase of 25-30k wfpw, well below normal peak for a leading edge node). We anticipate that Intel's client PC products (starting likely from Novalake) are likely to be outsourced to TSMC N2 in 2027-28 timeframe, due to competitive reasons (AMD and Apple Macs moving to N2 in 2027).

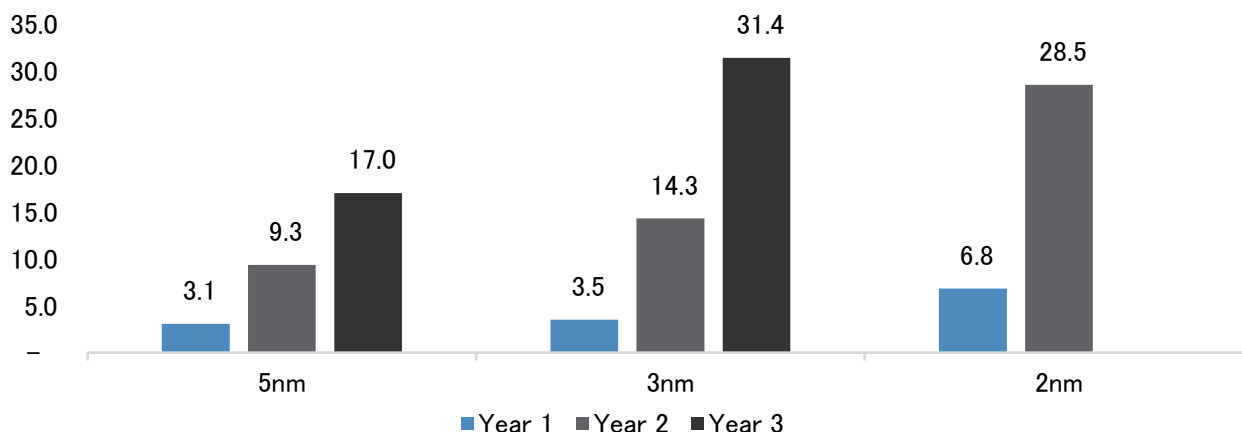
We continue to expect TSMC to build 20k wfpw / 60-65k wfpw by end 2025/2026. N2 wafer pricing is likely to come in at ~20-25% higher than N3E, with \$26-27k pricing for mobile versions of N2, while HPC versions with higher metal layer count are likely to be more expensive.

A faster ramp for N2, compared to N3, is expected, and should contribute \$6.8bn of revenues in 2026 (versus \$3.5bn of first year revenues for N3). Yield ramp for N2 also

appears quite smooth with initial chip probing yield already reaching 70% for early products.

Figure 4: TSMC revenue from N2, N3, and N5 in the first 3 years

US\$bn



Source: J.P. Morgan estimates, Company data.

Near-term eyes largely on tariffs and the degree of weakness in 4Q

In the earnings call, TSMC raised 2025 full year guidance to 30% yoy revenue growth in USD terms, from previous guidance of mid- 20s. While we've already highlighted the potential guidance raise ([link](#)), the magnitude turns out to be slightly above our expectation of high-20s.

Investor attention is likely to be largely focused on the implied qoq weakness (around 10% qoq revenue decline) in 4Q, which is weaker than Consensus and previous JPMe of 1%/6% qoq decline. We believe the implied weakness is mainly due to 1) high base in 3Q (rev guidance above market expectations), 2) TSMC taking a conservative view on tariffs (Sec 232 and country-specific tariffs) and not factoring in any exemptions for TSMC or Taiwan-made products at this point, and 3) TSMC also appears to be anticipating a marked slowdown in consumer demand as a result (including iPhone, in our view).

However, we feel that such a sharp slowdown in demand may not materialize, given that tariff outcomes so far have been better than feared. While we do believe that there has been some pull-in demand, especially in consumer electronics, in 1H25 (1H rev up 40% yoy in USD terms), but the extent is likely small given the ongoing tightness in N5 and N3 capacity since 2H24.

We therefore maintain our 6% qoq decline expectation for 4Q, and expect TSMC to eventually finish the year with low-mid 30% yoy growth in USD terms (JPMe 32%).

Capex outlook and Dividend growth

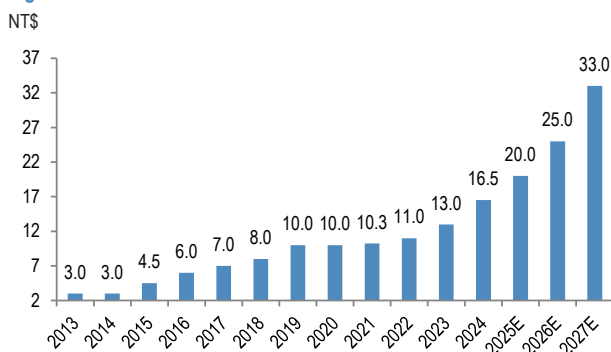
TSMC maintained its 2025 Capex guidance unchanged at US\$38-42bn (up ~30% yoy). While there's no further guidance for coming years yet, management only indicated no significant drop in Capex expected, despite very strong N2 demand, which is a bit more guarded, compared to investors' expectation of capex increase in 2026.

However, we still anticipate continued Capex growth to US\$45bn (up 13% yoy) in 2026, and maintained at a high level in 2027 (JPMe~US\$40bn) driven by 1) strong N2 demand, 2) continued strength in N3, and 3) continued accelerated buildout of capacity in the US (tools move in for the 2nd AZ Fab and construction of 3rd and 4th AZ Fab in 2026). In addition, we expect N2 demand to be upsized in case Intel signs a capacity agreement with TSMC for large-scale N2 outsourcing.

While the Capex seems to be high in terms of dollar amount, management also emphasized that sales growth would outgrow the capex growth. We anticipate capital intensity likely to be down to ~30% in 2027 versus 40-50% in 2023-25.

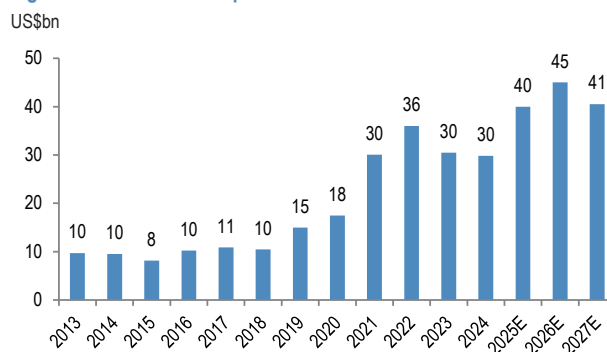
Meanwhile, we believe that TSMC's cash dividends should grow steadily due to strong FCF generation, and expect a 2023-27 DPS CAGR of 26%, with 2027E dividends reaching NT\$33 levels.

Figure 5: TSMC annual DPS



Source: J.P. Morgan estimates, Company data.

Figure 6: TSMC annual capex



Source: J.P. Morgan estimates, Company data.

2Q25 earnings review

Table 1: 2Q25 Earnings Review

NT\$bn, EPS in NT\$, %

NT\$ billion	2Q25			Variance		Growth	
	Actual	JPMe	BBG 28 days	vs. JPMe	vs. Consensus	Q/Q	Y/Y
Revenues	934	923	926	1%	1%	11%	39%
Gross profit	547	534	535	2%	2%	11%	53%
GM (%)	58.6%	57.9%	57.7%	74 bps	90 bps	-17 bps	545 bps
Operating profit	463	434	440	7%	5%	14%	62%
OPM (%)	49.6%	47.1%	47.5%	255 bps	211 bps	112 bps	708 bps
Net profit	398	368	376	8%	6%	10%	61%
EPS (NT\$)	15.36	14.20	14.49	8%	6%	10%	61%

Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates.

Forecast changes

Table 2: TSMC: Quarterly estimate revisions

NT\$ billion	Revised			Prior			Change (%)		
	3Q25E	4Q25E	1Q26E	3Q25E	4Q25E	1Q26E	3Q25E	4Q25E	1Q26E
Revenues (US\$ in mn)	32,781	30,871	30,688	31,233	29,251	29,800	5%	6%	3%
Revenues (NT\$ in bn)	951	895	890	906	848	864	5%	6%	3%
Gross profit	539	497	505	506	470	493	7%	6%	2%
GM (%)	56.7%	55.5%	56.7%	55.9%	55.4%	57.1%	84bps	4bps	-36bps
Operating profit	444	396	411	408	375	403	9%	6%	2%
OPM (%)	46.7%	44.3%	46.2%	45.1%	44.2%	46.6%	164bps	4bps	-36bps
Net profit	399	357	360	369	340	355	8%	5%	1%
EPS (NT\$)	15.39	13.75	13.88	14.23	13.11	13.68	8%	5%	1%
Operating Assumptions									
Capacity ('000 wafers, 12" equ)	3,909	3,974	4,024	3,909	3,974	4,024	0%	0%	0%
Shipment ('000 wafers, 12" equ)	3,614	3,365	3,322	3,549	3,276	3,313	2%	3%	0%
Shipment utilization (%)	92%	85%	83%	91%	82%	82%	167bps	223bps	23bps
Wafer ASP (US\$)	7,627	7,687	7,685	7,331	7,337	7,547	4%	5%	2%

Source: J.P. Morgan estimates.

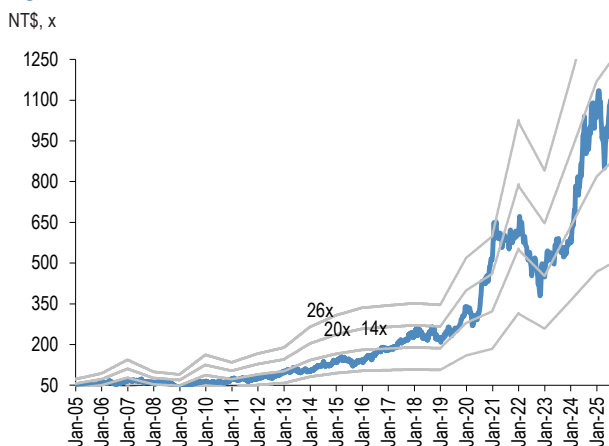
Table 3: TSMC: Annual estimate revisions

NT\$ billion	Revised			Prior			Change (%)		
	2025E	2026E	2027E	2025E	2026E	2027E	2025E	2026E	2027E
Revenues (US\$ in mn)	119,217	142,003	159,820	115,890	138,294	156,178	3%	3%	2%
Revenues (NT\$ in bn)	3,619	4,118	4,635	3,516	4,011	4,529	3%	3%	2%
Gross profit	2,077	2,361	2,663	2,004	2,281	2,569	4%	3%	4%
GM (%)	57.4%	57.3%	57.5%	57.0%	56.9%	56.7%	39bps	44bps	75bps
Operating profit	1,711	1,942	2,175	1,625	1,857	2,083	5%	5%	4%
OPM (%)	47.3%	47.2%	46.9%	46.2%	46.3%	46.0%	106bps	85bps	93bps
Net profit	1,515	1,688	1,902	1,439	1,626	1,831	5%	4%	4%
EPS (NT\$)	58.44	65.11	73.35	55.47	62.68	70.61	5%	4%	4%
Operating Assumptions									
Capacity ('000 wafers, 12" equ)	15,576	16,371	17,241	15,576	16,371	17,241	0%	0%	0%
Shipment ('000 wafers, 12" equ)	13,957	14,344	14,667	13,570	14,257	14,640	3%	1%	0%
Shipment utilization (%)	90%	88%	85%	87%	87%	85%	248bps	53bps	15bps
Wafer ASP (US\$)	7,207	8,165	8,978	7,152	7,981	8,858	1%	2%	1%

Source: J.P. Morgan estimates.

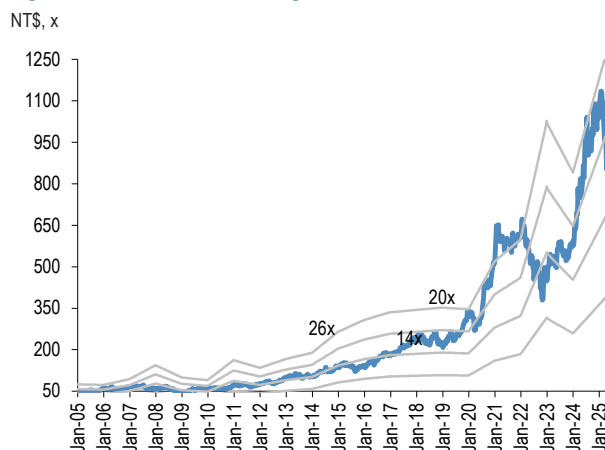
PE/PB charts

Figure 7: TSMC: 12-month forward P/E



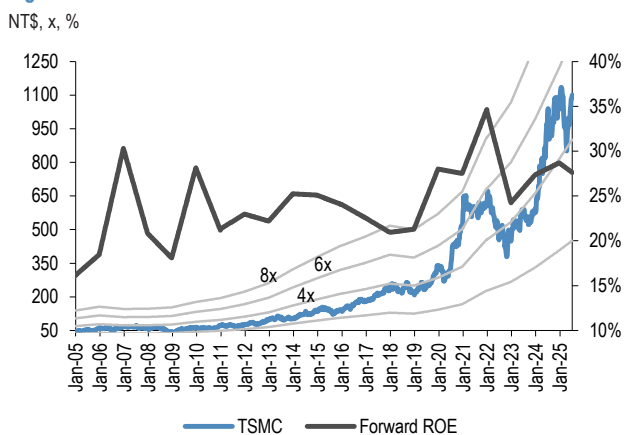
Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 8: TSMC: 12-month trailing P/E



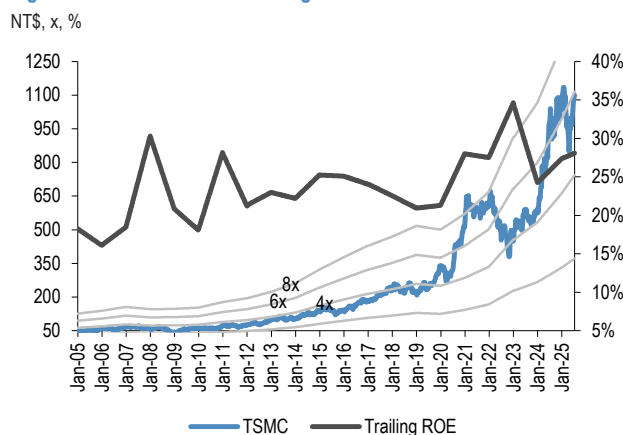
Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 9: TSMC: 12-month forward P/B and ROE



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 10: TSMC: 12-month trailing P/B and ROE



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

2Q25 Earnings Call highlights

2Q25 Earnings Call highlights

2Q25 recap

- Revenue up 17.8% QoQ in USD terms, supported by strong demand for N3/N5, partially offset by unfavorable FX
- GM 58.6%, down 0.2ppt, due to unfavorable FX and margin dilution from overseas fabs, partly balanced by higher-than-expected capacity utilization and cost improvement efforts
- 2Q25 main driver remains robust AI and HPC demand

Revenue mix by process node

- N3: 24% of sales
- N5: 36% of sales
- N7: 14% of sales

Revenue mix by platform

- HPC up 14% QoQ (60% of sales)
- Smartphones up 7% QoQ (27% of sales)

3Q25 outlook

- Revenue US\$31.8-33.0bn, up 8% QoQ at midpoint
- 3Q25 will be driven by leading edge process
- GM 55.5-57.5%, due to continued unfavored FX (NT\$32.88 in 1Q25 vs. NT\$31.05 in 2Q25) and more pronounced dilution from overseas fabs in Kumamoto and Arizona
- FX NT\$29
- OPM 45.5-47.5%
- Maintain 2025 capex to be US\$38-42bn

Overseas dilution

- Overseas dilution to continue in the next five years, starting from 2025
- GM dilution from overseas to begin with 2-3ppts in early stage and 3-4ppts in later stage
- TSMC will leverage increasing size in AZ to improve cost structure

FX impact

- Nearly all revenue is in USD, while ~75% of COGS is sold in NTD
- Sensitivity of revenue to the USD/TWD exchange rate is ~100%
- 1% appreciation in NTD vs. USD will reduce reported NTD revenue by 1% and GM by 0.4ppt

- 2Q25 revenue was negatively impacted by 4.4% and GM by 1.8ppts
- Assuming the USD/TWD rate to be NT\$29 in 3Q25, revenue will be down by 6.6% and GM down by 2.6ppts
- TSMC believes a long-term GM of 53% remains achievable by focusing on the fundamentals of its business and other factors for profitability

2025 outlook

- Looking into 2H25, TSMC hasn't seen any changes from customer behavior so far
- It understands the risk from potential tariff impact, especially on consumer electronics and price sensitive products
- It observes rebate program in China as short-term demand upside
- It expects a mild recovery in the overall non-AI end market
- Increasing AI model usage and adoption means more computation and leading-edge silicon. AI demand continues to be strong, including sovereign AI
- TSMC revises up its FY2025 revenue guidance to grow by ~30% in USD terms (vs. previous mid-20%) on strong demand for N3/N5 and growth in HPC
- It remains mindful of tariff impact and prudent in business planning in 2H25/2026 and will continue to invest for the future mega trend

US Fab update

- 6 advanced wafer plants, 2 advanced packaging fabs, and 1 R&D center
- 1st Fab (N4): MP in 4Q24
- 2nd Fab (N3): construction in completed, seeing strong interest from US customers and working to speed up the schedule by several quarters
- 3rdFab (N2/A16): construction has already begun and it is working on speeding up as well
- 4th Fab (N2/A16)
- 5th and 6th Fab will be on even more advanced nodes. Construction schedule will be based on customer needs
- 30% N2 and more advanced capacity will be in AZ upon completion

Japan Fab update

- 1st Fab: MP in late 2024 with good yield
- 2nd Fab: construction will start in late 2025

EU Fab update

- On track to build a specialty fab, ramp up schedule will be based on customer demand and market conditions

TW Fab update

- Plan to build 11 fabs and 4 advanced packaging fabs in Kaohsiung and Hsinchu in the next few years

N2/A16

- N2 new tape-out number in the first 2 years will be higher vs. N3/N5, fueled by smartphones and HPC
- N2 - 10-15% speed improvement at same power, 20-30% power improvement at same speed, and 15+% chip density increase vs. N3E (MP to start in 2H25, similar to N3 schedule)
- N2P - featuring better performance and power (MP in 2H26)
- A16 - with best-in-class superpower rail, SPR, providing 8-10% speed improvement at same power, 15-20% power improvement at same speed, 7-10% chip density increase vs. N2P, suitable for specific HPC and dense power delivery network (MP in 2H26)
- N2, N2P, A16 will be large and long-lasting nodes for TSMC

A14

- Will be in 2nd gen nano-sheet transistor structure
- A14 - 10-15% speed improvement at same power, 20-30% power improvement at same speed, and 20+% chip density increase vs. N2 (MP in 2028)
- Device performance and yield improvements are on or ahead of schedule
- SPR is planned for 2029

Q&A

Demand – AI demand looks better than 3 months ago, you mentioned CoWoS supply demand might be balanced by 2026. Is this still our view? Or do you think the capacity starts to look tighter?

- AI continues to be strong, so does CoWoS
- We are in a mode of narrowing the gap. The momentum is still there and very healthy.

On-device AI – are you seeing more development? Is it better than 3-6 months back?

- It takes 1-2 years to complete a new design on a product. The momentum is still going.
- The edge AI number is mild, and the die size will increase by 5-10% . Need to wait for around 6 months to 1 year to see an explosion

4Q25 revenue looks like it will decline. Is that based on your customers? Or is it TSMC being cautious in terms of the guidance?

- The comment of “TSMC becomes more conservative” – this is more real.
- It considers possible impact of tariff and other uncertainties - we become more conservative.
- CC guaranteed that with technology leadership and excellent execution, it will grab opportunities and achieve the high end of the target

GM – Can we reflect or earn our value from including the impact effects into the pricing? What is our confidence level on GM for next year? Can it maintain a similar level as this year?

- FX impact is huge. TSMC is working on selling our value. We are confident about 53% GM and higher - pay more attention to higher

H20 – H20 seems to be shipped to China, can our long term AI accelerator growth of mid 40% CAGR be higher?

- TSMC didn't see the signal. It is too early to give an estimate now.
- This is really positive for TSMC and its customers
- TSMC is not ready to increase its forecast, so it's probably more appropriate to answer this in another quarter.

N2 ramp – What will be the revenue contribution next year? It's surprising that N2 ramp is similar to N3. Can we expect 15% revenue contribution from N2 in 2026 and 10-11% in 2027?

- TSMC ramps up new node with smartphone being the first application. Now, it's not only smartphones but HPC products.
- Ramping profile is similar to N3 due to limitation of our capacity
- Revenue contribution will be bigger – N2 should not have the same price as N3

N5/N3 – the supply and demand for N5/N3. N5 conversion seems to be down, as well as N3/N5 conversion. But we don't see capacity expansion on N3. Does that mean N5 will be less utilized? Or should we try to build more N3 in the future?

- N3 capacity is very tight. It will continue for a couple of years
- N5 is also very tight – because some AI products still use N4, and will transition to N3 in the next 2 years
- TSMC's advantage is that it has a giga Fab cluster. We have 85-90% common tools for each node
- It uses N7 to support N5, converting N5 to N3. Our leading edge capacities are all very tight

GM – To what extent can TSMC adjust wafer pricing higher to neutralize cost increases in your business?

- When FX is against TSMC, it can lean on other factors (the 6 factors for the profitability) to mitigate the negative impact
- Raising the price is only one of the factors that it can leverage
- It believes GM 53% and higher is still achievable

AI in factories – How much economic benefit can TSMC get from applying AI to the fabs? Any gains on the cost structure?

- It uses AI in operations, manufacturing and R&D
- 1% productivity gains in a company of its size equals to US\$1bn gains

11 fabs in Taiwan – do we have enough capacity to support the strong demand to next year, and also to N2, and will we further convert more N5 to N3? How do you see the N2 capacity plan for 2026?

- TSMC saw lots of announcements on AI datacenters around the world, and demand for N2, N3, and N5 is very high

- It tries very hard to narrow the gap of demand and supply

N2 – What is the ROI of N2 vs. N3? The capex is higher, and the company yield on N2 is also pretty good, what is the development for N2?

- N2 profitability is better than N3
- N3 – it takes longer to catch up with the company average
- N2 – it should be back to the past level
- In the past, the corporate average GM was 50% and now it is 53%. So it's less meaningful to talk about how much time it takes to catch up with the corporate average
- N2 development is on track, ramping in 2H25. Revenue will come up in 1H26

Capex – 2025 guidance is stronger on AI demand, but we stay more conservative on the capex? Is it because of macro uncertainty or any constraint in construction? What is the capex outlook for 2026/27?

- Capex is based on the following year's business
- TSMC is mindful of all macro uncertainties when considering capex plans
- It is too early to talk about 2026, but it's unlikely to see sudden drop in capex dollar amounts in the following years

Cloud AI growth – What is the AI accelerator revenue growth we expect in 2025? What is our CoWoS capacity expansion plan for 2025? What is the plan for CoWoS capacity in 2026?

- It tries to narrow the gap for now and 2026
- Demand is strong and healthy, and it's building new facilities in the back-end to increase CoWoS capacity
- AI demand is really strong, so is CoWoS capacity

Advanced packaging – The die size is getting bigger, requiring more power consumption and energy efficiency. What is our strategy for advanced packaging along with advanced node development? Any solutions we are prioritizing? What about the timeline and road map?

- Timeline is based on customers. It develops technology and increases capacity when the customer has such demand
- For advanced packaging, customers have different approaches. So we develop different technologies for all the customers, which is related to leading edge technology.
- There are some similarities in the various technologies, or it will take too much effort and get no return

Mature node – There is overcapacity industry-wide for the older node. Can we consolidate different nodes and how do we protect our profitability?

- TSMC develops specialty for older node (e.g. RF technology, CMOS, or high voltage)
- It is not worried about overcapacity, otherwise it will not build fabs in Germany and Japan. It's all related to customer demand

Humanoid robot – We start to see some contributions from robots, how does TSMC evaluate the market size? What are the opportunities in the long run for leading edge and mature node?

- It's too early to talk about it in 2025. Humanoid is too complicated, it will be used in the medical industry most of the time
- A lot of sensor tech. It is complex because it deals with human directly
- Its customer think EV cars are nothing, as robots will be 10 times of EV

Pricing – Do we see customers pulling in earlier than in 2026? More color on 4Q25?

- No. TSMC hasn't seen any different customer behavior so far.
- N3 cycle time is 4 months - there's no way to pull in anything, and the capacity is really tight. We have all the schedules and little room for pulling

Capital intensity – When investing in a new node, we see capital intensity jumping to >40%. We see strong demand for N2 for multiple years, what is your expectation on capital intensity?

- Capex is invested in the future growth opportunities
- If TSMC does its job well, we should see higher growth in revenue than the growth in capex, where there shouldn't be high capital intensity

AI adoption of leading edge node – how do we see the AI adoption of the most leading edge node for TSMC? How do we see that going forward with A16?

- HPC customers are usually one step behind, usually N+1 or N+2
- Power consumption is very important. Data center suppliers build electricity plants to support their AI data centers

Overseas expansion – TSMC is speeding up its US 2nd fab, noting the US ITC bill. How does this impact our US expansion and any implication on overseas dilution?

- The ramp-up schedule is based on customer demand. It appreciates the US government increasing the ITC from 25% to 35%
- Margin impact should be positive but not significant in the 5 year period

US investment – Does speeding up US investment impact other regional developments in Japan and Germany? What is the breakdown of overseas vs. domestic capex going forward?

- US Fab: for the leading-edge
- Japan Fab: for specialty nodes, specifically for CMOS sensors
- Germany Fab: for the automotive industry
- They are not in the same field, so it should not affect fab investment in Japan and Germany

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93.3	93.8	93.8	93.8	93.8	93.8	93.8	93.8	93.8	93.8
30.974	30.974	32.781	30.971	30.971	30.971	30.971	30.971	30.971	30.971
-386.4	-386.4	-411.4	-398.7	-398.7	-398.7	-398.7	-398.7	-398.7	-398.7
15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5	15.5
547.4	547.4	539.2	498.6	498.6	498.6	498.6	498.6	498.6	498.6
83.9	83.9	-95.1	-100.3	-100.3	-100.3	-100.3	-100.3	-100.3	-100.3
-22.7	-22.7	-23.8	-26.8	-26.8	-26.8	-26.8	-26.8	-26.8	-26.8
61.3	61.3	-71.3	-71.6	-71.6	-71.6	-71.6	-71.6	-71.6	-71.6
463.4	463.4	444.1	396.3	396.3	396.3	396.3	396.3	396.3	396.3
21.5	21.5	20.6	21.2	21.2	21.2	21.2	21.2	21.2	21.2
8.1	8.1	10	10	10	10	10	10	10	10
493.0	493.0	466.8	416.6	416.6	416.6	416.6	416.6	416.6	416.6
-56.5	-56.5	-67.5	-67.8	-67.8	-67.8	-67.8	-67.8	-67.8	-67.8
398.3	398.3	399.0	356.6	356.6	356.6	356.6	356.6	356.6	356.6
25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9
15.36	15.36	15.39	13.75	13.75	13.75	13.75	13.75	13.75	13.75
6.841	6.841	7.627	7.687	7.687	7.687	7.687	7.687	7.687	7.687
96	96	92	85	85	85	85	85	85	85
3.774	3.774	3.909	3.974	3.974	3.974	3.974	3.974	3.974	3.974
3.718	3.718	3.614	3.365	3.365	3.365	3.365	3.365	3.365	3.365
94	94	92	85	85	85	85	85	85	85
58.6	58.6	56.7	55.5	55.5	55.5	55.5	55.5	55.5	55.5
49.6	49.6	46.7	44.3	44.3	44.3	44.3	44.3	44.3	44.3
63.8	63.8	66.5	65.4	65.4	65.4	65.4	65.4	65.4	65.4
69	69	61	42.0	39.8	39.8	39.8	39.8	39.8	39.8
8	8	19	8	-2	-2	-2	-2	-2	-2
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Investment Thesis, Valuation and Risks

TSMC (*Overweight; Price Target: NT\$1275.0*)

Investment Thesis

We expect TSMC's structural growth drivers to remain strong due to its near-monopoly position in AI accelerators and edge AI, supported by its robust process roadmap (N3 and N2) and industry-leading packaging technology. We believe TSMC could grow low-mid 30% in USD terms in 2025 and healthy growth in 2026, driven by broader AI demand pick-up (e.g. CSPs, sovereigns and China demand) with CoWoS capacity remaining tight into 2026. Additionally, we anticipate the GM to rebound in 2026 with price hikes, better N3 yield, and accelerated productivity and cost-down measures.

Valuation

Our Dec-25 PT of NT\$1,275 is based on ~20x 12-month forward P/E and reflects the positive fundamental demand driver, continued growth in N3 and stronger N2 ramp in 2026. Our target multiple is higher than TSMC's five-year average historical multiple.

Risks to Rating and Price Target

Key downside risks to our rating and price target include: (1) uncertainty regarding the impact of tariffs and a global economic slowdown; (2) worse-than-expected AI demand in 2026; and (3) greater GM dilution from US fabs due to the faster ramp.

TSMC: Summary of Financials

Income Statement - Annual					Income Statement - Quarterly				
	FY24A	FY25E	FY26E	FY27E		1Q25A	2Q25A	3Q25E	4Q25E
Revenue	2,894,308	3,618,960	4,118,088	4,634,792	Revenue	839,254	933,792	950,642	895,272
COGS	(1,269,954)	(1,542,424)	(1,757,443)	(1,971,580)	COGS	(345,859)	(386,423)	(411,429)	(398,714)
Gross profit	1,624,354	2,076,535	2,360,645	2,663,212	Gross profit	493,395	547,369	539,213	496,558
SG&A	(98,119)	(104,848)	(118,072)	(134,184)	SG&A	(29,767)	(22,666)	(23,766)	(28,649)
Adj. EBITDA	1,984,850	2,451,520	2,857,399	3,201,001	Adj. EBITDA	582,220	651,481	632,577	585,242
D&A	(662,797)	(740,580)	(915,088)	(1,025,983)	D&A	(175,139)	(188,058)	(188,428)	(188,954)
Adj. EBIT	1,322,053	1,710,940	1,942,311	2,175,018	Adj. EBIT	407,081	463,423	444,149	396,287
Net Interest	76,718	85,510	65,662	88,241	Net Interest	22,182	21,501	20,581	21,246
Adj. PBT	1,405,839	1,808,264	2,012,205	2,267,620	Adj. PBT	430,895	493,036	465,761	418,571
Tax	(233,407)	(296,027)	(326,871)	(368,565)	Tax	(70,163)	(95,543)	(67,535)	(62,786)
Minority Interest	836	3,171	3,120	3,120	Minority Interest	831	780	780	780
Adj. Net Income	1,173,268	1,515,408	1,688,455	1,902,176	Adj. Net Income	361,564	398,273	399,006	356,566
Reported EPS	45.24	58.44	65.11	73.35	Reported EPS	13.94	15.36	15.39	13.75
Adj. EPS	45.24	58.44	65.11	73.35	Adj. EPS	13.94	15.36	15.39	13.75
DPS	16.50	20.00	25.00	33.00					
Payout ratio	51.0%	44.2%	42.8%	50.7%					
Shares outstanding	25,933	25,933	25,933	25,933	Shares outstanding	25,933	25,933	25,933	25,933
Balance Sheet & Cash Flow Statement					Ratio Analysis				
	FY24A	FY25E	FY26E	FY27E		FY24A	FY25E	FY26E	FY27E
Cash and cash equivalents	2,422,020	2,402,455	3,041,525	3,916,622	Gross margin	56.1%	57.4%	57.3%	57.5%
Accounts receivable	272,088	412,070	530,469	553,095	EBITDA margin	68.6%	67.7%	69.4%	69.1%
Inventories	287,869	245,280	315,755	329,223	EBIT margin	45.7%	47.3%	47.2%	46.9%
Other current assets	106,376	32,555	41,909	43,697	Net profit margin	40.5%	41.9%	41.0%	41.0%
Current assets	3,088,352	3,092,361	3,929,658	4,842,637	ROE	30.3%	31.7%	29.1%	27.8%
PP&E	3,234,980	3,719,215	4,109,127	4,257,644	ROA	19.2%	21.9%	21.7%	21.3%
LT investments	149,040	139,496	143,728	148,089	ROCE	22.7%	24.8%	24.1%	23.4%
Other non current assets	219,565	217,799	217,799	217,799	SG&A/Sales	3.4%	2.9%	2.9%	2.9%
Total assets	6,691,938	7,168,871	8,400,313	9,466,169	Net debt/equity	NM	NM	NM	NM
Short term borrowings	59,858	94,214	94,214	94,214	P/E (x)	25.0	19.3	17.4	15.4
Payables	74,227	98,112	126,302	131,689	P/BV (x)	6.8	5.5	4.6	4.0
Other short term liabilities	1,130,441	663,164	853,709	890,122	EV/EBITDA (x)	14.1	11.4	9.5	8.2
Current liabilities	1,264,525	855,490	1,074,225	1,116,026	Dividend Yield	1.5%	1.8%	2.2%	2.9%
Long-term debt	958,429	848,535	848,535	848,535	Sales/Assets (x)	47.4%	52.2%	52.9%	51.9%
Other long term liabilities	145,408	147,892	120,459	98,114	Interest cover (x)	NM	NM	NM	NM
Total liabilities	2,368,362	1,851,917	2,043,219	2,062,675	Operating leverage	128.3%	117.5%	98.0%	95.5%
Shareholders' equity	4,288,545	5,281,396	6,321,536	7,367,936	Revenue y/y Growth	33.9%	25.0%	13.8%	12.5%
Minority interests	35,031	35,558	35,558	35,558	EBITDA y/y Growth	36.5%	23.5%	16.6%	12.0%
Total liabilities & equity	6,691,938	7,168,871	8,400,313	9,466,169	Tax rate	16.6%	16.4%	16.2%	16.3%
BVPS	165.4	203.7	243.8	284.1	Adj. Net Income y/y Growth	39.9%	29.2%	11.4%	12.7%
y/y Growth	24.0%	23.2%	19.7%	16.6%	EPS y/y Growth	39.9%	29.2%	11.4%	12.7%
Net debt/(cash)	(1,403,733)	(1,459,706)	(2,098,776)	(2,973,873)	DPS y/y Growth	26.9%	21.2%	25.0%	32.0%
Cash flow from operating activities	1,975,661	1,785,853	2,620,930	2,928,958					
o/w Depreciation & amortization	662,797	740,580	915,088	1,025,983					
o/w Changes in working capital	(90,016)	(73,508)	(160,684)	(30,707)					
Cash flow from investing activities	(928,044)	(1,213,504)	(1,309,232)	(1,178,861)					
o/w Capital expenditure	(958,198)	0	0	0					
as % of sales	33.1%	0.0%	0.0%	0.0%					
Cash flow from financing activities	(313,242)	(591,913)	(672,628)	(875,000)					
o/w Dividends paid	(428,731)	(521,824)	(651,435)	(858,896)					
o/w Net debt issued/(repaid)	90,711	(75,538)	0	0					
Net change in cash	734,375	(19,564)	639,069	875,097					
Adj. Free cash flow to firm	1,142,360	561,038	1,315,930	1,754,458					
y/y Growth	164.7%	(50.9%)	134.6%	33.3%					

Source: Company reports and J.P. Morgan estimates.

Note: NT\$ in millions (except per-share data). Fiscal year ends Dec. o/w - out of which

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TSMC (2330.TW, 2330 TT) Price Chart



Date	Rating	Price (NT\$)	Price Target (NT\$)
11-Oct-22	OW	438.0	650
20-Oct-23	OW	546.0	700
04-Jan-24	OW	578.0	750
19-Jan-24	OW	588.0	770
06-Mar-24	OW	730.0	850
07-Apr-24	OW	780.0	900
17-Jun-24	OW	922.0	1,080
19-Jul-24	OW	1005.0	1,200
18-Oct-24	OW	1035.0	1,500
13-Apr-25	OW	889.0	1,300
18-Apr-25	OW	847.0	1,275

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 28, 2002. All share prices are as of market close on the previous business day.

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